



Time Series Analysis

Seminar 4: Seasonality and forecasting: SARIMA, TBATS, Prophet



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Acronyms used in this material

- ▣ ACF: Autocorrelation Function
- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ ARMA: AutoRegressive Moving Average (model)
- ▣ DHR: Dynamic Harmonic Regression
- ▣ DM: Diebold–Mariano (test)
- ▣ ENTSO-E: European Network of Transmission System Operators for Electricity
- ▣ GDP: Gross Domestic Product
- ▣ GW: Gigawatt
- ▣ HICP: Harmonised Index of Consumer Prices
- ▣ HLN: Harvey–Leybourne–Newbold (small-sample correction of the DM test)
- ▣ MA: Moving Average
- ▣ MAE: Mean Absolute Error
- ▣ MAPE: Mean Absolute Percentage Error
- ▣ MASE: Mean Absolute Scaled Error
- ▣ MSE: Mean Squared Error
- ▣ MW: Megawatt
- ▣ PACF: Partial AutoCorrelation Function
- ▣ RMSE: Root Mean Squared Error
- ▣ SARIMA: Seasonal ARIMA (model)
- ▣ SCA: Seasonally and Calendar Adjusted
- ▣ TBATS: Trigonometric seasonality, Box–Cox, ARMA errors, Trend, Seasonal components (model)
- ▣ WD: Working Days (regressor)

Today's question and route

- **Question:** how do we model, forecast and compare forecasts of series that repeat a pattern every year, week or day?
 - ▶ this seminar comes **before** Lecture 4: the section “Prerequisites for today” gives every formula the tasks use
- **Route**
 - ▶ Part A: seasonal differences, SARIMA polynomials and autocorrelations, the Diebold–Mariano test and forecast combination on paper
 - ▶ Part B: Romanian GDP, industrial production, daily electricity load and monthly inflation
 - ▶ Part C: tourism after the pandemic and an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1, A2	seasonal differences and seasonal naive forecasts; SARIMA polynomials	Solved, Proposed	A1
A3, A4	the ACF of the airline model; a seasonal AR and moment estimates	Solved, Proposed	A3
A5, A6	MASE and the Diebold–Mariano test; combining two forecasts	Solved, Proposed	A5
B1, B2	SARIMA: Romanian GDP; industrial production with working days	Solved, Proposed	B1
B3, B4	cross-validation: electricity load; monthly inflation	Solved, Proposed	B3
C1, C2	tourism after the pandemic; what is wrong in an AI answer?	Proposed	B1, A5

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
Real GDP, not adjusted	Eurostat (namq_10_gdp)	quarterly	2000–2026
Industrial production	Eurostat (sts_inpr_m)	monthly, 2021 = 100	2010–2026
Electricity load	ENTSO-E	daily mean of hourly data	2022–2026
HICP, Romania	Eurostat (prc_hicp_minr)	monthly, 2015 = 100	2010–2026
Tourism nights	Eurostat (tour_occ_nim)	monthly	2005–2025

- ▣ Logs times 100: $100 \ln Y_t$, so that differences are growth rates in %
- ▣ ENTSO-E: European Network of Transmission System Operators for Electricity; load = electricity consumed, in MW (hourly average)
- ▣ In the notebook: `fit_sarima`, `b1_gdp()`, `b3_load_cv()`; no account or key is needed

Prerequisites for today (1/5): seasonality and seasonal differences

- **Seasonality:** a pattern that repeats with a fixed **period** s : $s = 4$ (quarters), $s = 12$ (months), $s = 7$ (days of the week), $s = 24$ (hours of the day)
 - ▶ causes: weather, the calendar (working days, holidays), institutions (tax dates, school year)
- **Seasonal difference:** $\Delta_s y_t = (1 - L^s)y_t = y_t - y_{t-s}$; with logs, $100 \Delta_4 \ln Y_t$ is the year-on-year growth rate
 - ▶ L : the lag operator, $L^s y_t = y_{t-s}$; $\Delta_s y_t$ compares each value with the same season of the previous year
 - ▶ both differences: $\Delta \Delta_s y_t = (1 - L)(1 - L^s)y_t = y_t - y_{t-1} - y_{t-s} + y_{t-s-1}$
- **Seasonal naive forecast:** $\hat{y}_{T+h} = y_{T+h-s}$ for $h \leq s$: the value of the same season one period earlier ([Chapter 0](#))
 - ▶ the benchmark every seasonal model must beat

Prerequisites for today (2/5): SARIMA models

- **SARIMA**(p, d, q)(P, D, Q) $_s$: $\phi(L)\Phi(L^s)(1-L)^d(1-L^s)^D y_t = \theta(L)\Theta(L^s)\varepsilon_t$
 - ▶ $\phi(L)$, $\theta(L)$: the regular AR and MA polynomials (orders p , q , [Chapter 2](#)); d , D : the numbers of regular and seasonal differences
 - ▶ $\Phi(L^s) = 1 - \phi_1 L^s - \dots - \phi_P L^{Ps}$ (seasonal AR), $\Theta(L^s) = 1 + \theta_1 L^s + \dots + \theta_Q L^{Qs}$ (seasonal MA)
 - ▶ the polynomials are **multiplied**: cross terms such as $\theta\Theta L^{s+1}$ appear without new parameters
- **Airline model** (Box and Jenkins): SARIMA(0, 1, 1)(0, 1, 1) $_s$
 - ▶ $w_t = \Delta\Delta_s y_t = (1 + \theta L)(1 + \Theta L^s)\varepsilon_t$: two parameters, θ and Θ
 - ▶ ACF of w_t : $\rho_1 = \frac{\theta}{1+\theta^2}$, $\rho_s = \frac{\Theta}{1+\Theta^2}$, $\rho_{s-1} = \rho_{s+1} = \rho_1\rho_s$, all other lags 0
- **Identification** at seasonal lags $s, 2s, 3s$: as in [Chapter 2](#), but one season apart
 - ▶ ACF cuts off after lag s , PACF decays at $s, 2s, \dots$: seasonal MA(1); the reverse: seasonal AR(1)

Prerequisites for today (3/5): calendar regressors and Fourier terms

- **Regression with SARIMA errors:** $y_t = \beta^\top x_t + u_t$, $u_t \sim \text{SARIMA}$; x_t known in advance: working days, holidays, outliers
 - ▶ x_t : the vector of regressors; β : their coefficients; u_t : the error, with SARIMA dynamics
 - ▶ **working days:** number of Monday–Friday days of the month that are not public holidays
 - ▶ with $100 \ln y_t$, the coefficient of a dummy is approximately the effect in %
- **Fourier terms** for a period m : $\sin(2\pi kt/m)$, $\cos(2\pi kt/m)$, $k = 1, \dots, K$
 - ▶ k : the harmonic, a wave with k cycles per period; K : the number of harmonics used
 - ▶ $2K$ coefficients instead of $m - 1$ dummies; m may be non-integer (365.25 days in a year)
 - ▶ **dynamic harmonic regression (DHR):** Fourier terms plus calendar dummies, with ARMA errors
- TBATS and Prophet (Lecture 4) are two automatic versions of the same idea: several seasonal periods, Fourier-type seasonality

Prerequisites for today (4/5): comparing forecasts

- **Time-series cross-validation:** origins $T_1 < T_2 < \dots$; at each origin, fit on the data up to it and forecast h steps
 - ▶ e_i : the n forecast errors collected over all origins and horizons
 - ▶ $\text{MAE} = \frac{1}{n} \sum |e_i|$; $\text{RMSE} = \sqrt{\frac{1}{n} \sum e_i^2}$; $\text{MASE} = \text{MAE} / (\text{in-sample MAE of the seasonal naive method})$ ([Chapter 0](#))
- **Diebold–Mariano (DM) test of equal accuracy:** $d_t = L(e_{1t}) - L(e_{2t})$, loss $L(e) = e^2$ or $|e|$
 - ▶ e_{1t}, e_{2t} : the errors of forecasts 1 and 2; d_t : the loss differential; $\bar{d}$: its mean over the n values
 - ▶ $\text{DM} = \bar{d} / \sqrt{\hat{\sigma}_d^2 / n}$, $\hat{\sigma}_d^2 = \frac{1}{n} \sum (d_t - \bar{d})^2$ for one-step (or non-overlapping) losses
 - ▶ small-sample correction (Harvey, Leybourne, Newbold): $\text{HLN} = \sqrt{(n-1)/n} \text{DM}$, compared with Student $t(n-1)$
 - ▶ $\bar{d} < 0$: forecast 1 is more accurate; 5% critical values $t_{0.975}(5) = 2.571$, $t_{0.975}(7) = 2.365$

Prerequisites for today (5/5): forecast combination

□ **Forecast combination:** $f_c = wf_1 + (1 - w)f_2$; for unbiased forecasts

- ▶ f_1, f_2 : the two forecasts; w : the weight of f_1 ; σ_1, σ_2 : the standard deviations of their errors; ρ : the correlation of the errors
- ▶ $\text{MSE}(w) = w^2\sigma_1^2 + (1 - w)^2\sigma_2^2 + 2w(1 - w)\rho\sigma_1\sigma_2$; best $w^* = \frac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$
- ▶ MSE: the mean squared error; w^* minimises it; the gain over the better forecast is large when ρ is small

A1: seasonal differences by hand [Solved]

Task

- ▣ A quarterly series, $t = 1, \dots, 9$: 10, 14, 16, 18, 12, 16, 19, 21, 14.
- ▣ 1. Compute $\Delta_4 y_t$ for $t = 5, \dots, 9$.
- ▣ 2. Compute $\Delta \Delta_4 y_t$ for $t = 6, \dots, 9$.
- ▣ 3. Expand $(1 - L)(1 - L^4)$ and check it on $t = 9$.
- ▣ 4. Give the seasonal naive forecasts for $t = 10, \dots, 13$.
- ▣ Report: thirteen numbers and one sentence on what Δ_4 removes.

Solution

- ▣ 1. $\Delta_4 y$: 2, 2, 3, 3, 2 (e.g. $12 - 10$, $16 - 14$)
- ▣ 2. $\Delta \Delta_4 y$: 0, 1, 0, -1
- ▣ 3. $1 - L - L^4 + L^5$; $t = 9$: $14 - 21 - 12 + 18 = -1$
- ▣ 4. $\hat{y}_{10}, \dots, \hat{y}_{13} = 16, 19, 21, 14$ (the last four values)
- ▣ Δ_4 removes a seasonal pattern that repeats exactly; the small, stable $\Delta_4 y$ shows the year-on-year growth.

A2: SARIMA polynomials [Proposed]

Solution discussed in class

Task

- ▣ Monthly data, $s = 12$. Model: A1.
- ▣ 1. Write the airline model $\text{SARIMA}(0, 1, 1)(0, 1, 1)_{12}$ with $\theta = -0.4$, $\Theta = -0.6$ as an equation for y_t .
- ▣ 2. List the lags of y and of ε that appear, and the coefficient of ε_{t-13} .
- ▣ 3. Expand $(1 - 0.5L)(1 - 0.3L^{12})$ for $\text{SARIMA}(1, 0, 0)(1, 0, 0)_{12}$.
- ▣ 4. Count the parameters of $\text{SARIMA}(1, 1, 1)(1, 1, 1)_{12}$, including σ^2 .
- ▣ Report: two equations, two lists of lags and one count.

A3: the ACF of the airline model [Solved]

Task

- ▣ Quarterly airline model:
 $w_t = \Delta\Delta_4 y_t = (1 - 0.5L)(1 - 0.6L^4)\varepsilon_t$,
 $\varepsilon_t \sim \text{WN}(0, \sigma^2)$.
- ▣ 1. Write w_t as an MA(5) and list its nonzero coefficients.
- ▣ 2. Compute $\gamma(0)/\sigma^2$.
- ▣ 3. Compute $\rho_1, \dots, \rho_5$.
- ▣ 4. Say what the sample ACF of w_t should look like.
- ▣ Report: five autocorrelations and one sentence.

Solution

- ▣ 1. $w_t = \varepsilon_t - 0.5\varepsilon_{t-1} - 0.6\varepsilon_{t-4} + 0.3\varepsilon_{t-5}$
- ▣ 2. $\gamma(0)/\sigma^2 = 1 + 0.25 + 0.36 + 0.09 = (1 + 0.25)(1 + 0.36) = 1.70$
- ▣ 3. $\rho_1 = -0.5/1.25 = -0.400$; $\rho_4 = -0.6/1.36 = -0.441$;
 $\rho_3 = \rho_5 = \rho_1\rho_4 = 0.176$; $\rho_2 = 0$
- ▣ 4. Spikes at lags 1 and 4 with small “satellites” at 3 and 5; nothing after lag 5: the signature of the airline model.

A4: a seasonal AR and moment estimates [Proposed]

Solution discussed in class

Task

- Model: A3.
- 1. For $y_t = 0.7y_{t-4} + \varepsilon_t$, compute ρ_4 , ρ_8 , ρ_{12} and say which other lags are zero.
- 2. Describe its PACF.
- 3. The sample ACF of $\Delta\Delta_{12}y_t$ is -0.40 at lag 1, -0.45 at lag 12, small at lags 11 and 13, and inside the band elsewhere; propose a model.
- 4. Estimate θ and Θ from ρ_1 and ρ_{12} , keeping the invertible roots.
- Report: three autocorrelations, a model and two estimates.

A5: MASE and the Diebold–Mariano test [Solved]

Task

- ▣ Six non-overlapping forecast windows; errors of method 1: 0.4, −0.6, 0.9, −0.3, 0.5, −0.8; of method 2: 1.1, −0.9, 0.7, −1.2, 0.8, −1.0; the in-sample MAE of the seasonal naive method is 1.
- ▣ 1. Compute the MAE and the MASE of both methods.
- ▣ 2. Compute $d_t = e_{1t}^2 - e_{2t}^2$, $\bar{d}$ and $\hat{\sigma}_d^2$.
- ▣ 3. Compute DM and HLN.
- ▣ 4. Decide at 5% with $t(5)$.
- ▣ Report: four accuracy measures, the test statistic and a decision.

Solution

- ▣ 1. MAE: 0.583 and 0.950; MASE: the same (scale 1): both below 1, method 1 better
- ▣ 2. d : −1.05, −0.45, 0.32, −1.35, −0.39, −0.36;
 $\bar{d} = -0.547$; $\hat{\sigma}_d^2 = 0.2864$
- ▣ 3. $DM = -0.547 / \sqrt{0.2864/6} = -2.50$;
 $HLN = 0.913 \cdot (-2.50) = -2.28$
- ▣ 4. $|-2.28| < 2.571$ ($p = 0.071$): equal accuracy is not rejected; six windows are too few to prove the visible gain.

A6: two forecasts and their combination [Proposed]

Solution discussed in class

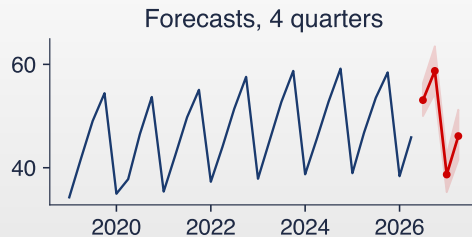
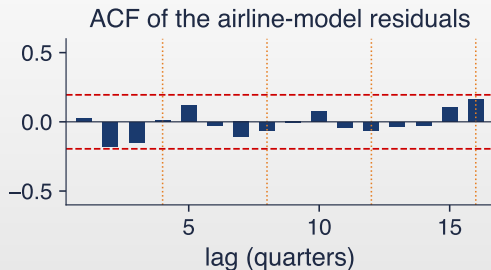
Task

- ▣ Eight windows; errors e_1 : 2.0, -1.5, 1.0, -2.5, 3.0, 0.5, -1.0, 1.5; e_2 : 1.0, -2.5, 2.0, -0.5, 1.5, 2.0, -2.0, 0.5. Model: A5.
- ▣ 1. Compute the MAE of both methods and the DM and HLN statistics with squared losses.
- ▣ 2. Decide at 5% with $t(7)$.
- ▣ 3. The equal-weight combination has errors $(e_1 + e_2)/2$: compute its MAE and RMSE.
- ▣ 4. For unbiased forecasts with $\sigma_1 = 2$, $\sigma_2 = 3$, $\rho = 0.3$, compute $\text{MSE}(0.5)$, w^* and $\text{MSE}(w^*)$.
- ▣ Report: six numbers, a decision and three numbers for the combination.

B1: the airline model for Romanian GDP [Solved]

- ▣ **Question:** how stable is the seasonal pattern of Romanian GDP, and what does the airline model forecast for the next four quarters?
- ▣ **Inputs:** Eurostat, real GDP not adjusted, 2000Q1–2026Q2, $T = 106$; $y_t = 100 \ln Y_t$
- ▣ **Tasks**
 1. Estimate SARIMA(0, 1, 1)(0, 1, 1)₄ by maximum likelihood and report $\hat{\theta}$, $\hat{\Theta}$ and their standard errors.
 2. Test the residuals with Ljung–Box $Q^*(8)$ on $8 - 2 = 6$ degrees of freedom.
 3. Forecast the next four quarters, in bn EUR, with 95% intervals.
 4. Compute the implied year-on-year growth for each forecast quarter.
 5. Interpretation: what does $\hat{\Theta}$ say about how fast the seasonal pattern of GDP changes?
- ▣ **Report:** two estimates with standard errors, a test, four forecasts with intervals and one sentence
- ▣ **Notebook:** section B1

B1: solution (1/2) [Solved]



----- $\pm 1.96/\sqrt{T}$

— GDP, not adjusted (bn EUR)

—•— forecast

95% interval

B1: solution (2/2) [Solved]

- $\hat{\theta} = -0.171$ (0.109): not significant; $\hat{\Theta} = -0.585$ (0.072); $\hat{\sigma} = 3.05\%$; $Q^*(8) = 9.30$, $p = 0.157$: no residual autocorrelation
- Forecasts from 2026Q3: 53.1, 58.7, 38.7, 46.1 bn EUR; first interval [50.0, 56.4], fourth [41.5, 51.2]
- Implied year-on-year growth: -0.84% , 0.52% , 0.83% , 0.53%
- Interpretation: the seasonal forecast is an exponentially weighted average of past seasonal patterns, with weight $1 + \hat{\Theta} = 0.42$ on the newest year; $\Theta = -1$ would mean a fixed pattern, $\Theta = 0$ a pattern that changes freely: the pattern of GDP evolves slowly

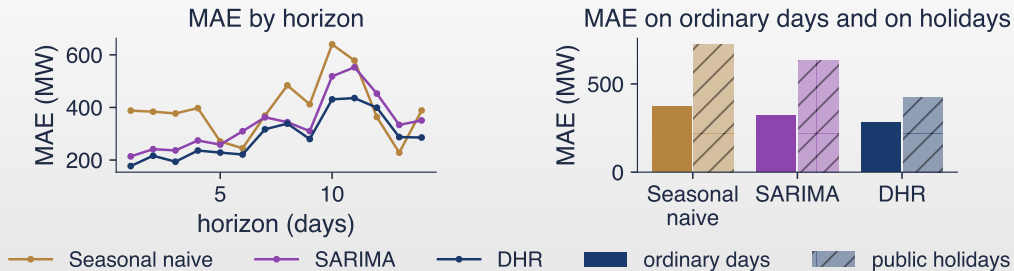
B2: industrial production and working days [Proposed]

- **Question:** does the number of working days explain part of the monthly movements of Romanian industrial production?
- **Inputs:** Eurostat, industrial production not adjusted, 2010–July 2026, $y_t = 100 \ln Y_t$; dummies for April and May 2020; model: B1
- **Tasks**
 1. Estimate the airline model with and without the number of working days as a regressor, and compare AICc and $Q^*(24)$.
 2. Add two alternatives with working days, $(2, 1, 0)(0, 1, 1)_{12}$ and $(1, 1, 1)(0, 1, 1)_{12}$, and choose a model.
 3. Report the working-day coefficient with its standard error, and forecast the next 12 months.
 4. Interpretation: why must the working-day effect be removed before two months are compared?
- **Report:** a table of AICc and p-values, one coefficient, the forecast chart and one sentence
- **Notebook:** section B2

B3: cross-validation for daily electricity load [Solved]

- **Question:** does a regression with Fourier terms and holiday dummies forecast daily load better than SARIMA and the seasonal naive method?
- **Inputs:** ENTSO-E, daily mean load in Romania (GW), 2022–2026; 13 origins every 28 days from 30 June 2025 to 1 June 2026, horizon 14 days
- **Tasks**
 1. At each origin, fit the seasonal naive method (period 7), SARIMA(1, 0, 1)(0, 1, 1)₇ and a DHR (4 annual Fourier pairs, weekday and holiday dummies, ARMA(1,1) errors).
 2. Compute the MAE and the MASE of each method over all origins and horizons.
 3. Test DHR against SARIMA with the DM test on the window-average squared errors (one value per origin), with the HLN correction.
 4. Compare the MAE on public holidays with the MAE on ordinary days.
 5. Interpretation: why is the gain of the DHR largest on public holidays?
- **Report:** a table of MAE and MASE, one test, the chart and one sentence
- **Notebook:** section B3

B3: solution (1/2) [Solved]



B3: solution (2/2) [Solved]

Method	MAE (MW)	MASE	holidays	other days
Seasonal naive	395	1.28	723	372
SARIMA	340	1.10	630	319
DHR	289	0.94	422	280

- MASE scale: in-sample MAE of the weekly seasonal naive method, 309 MW; DM (HLN), DHR against SARIMA: -1.96 , $p = 0.073$; against the seasonal naive method: -2.73 , $p = 0.018$
- Interpretation: SARIMA and the seasonal naive method only know “one week ago”; a holiday that falls on a weekday looks to them like a normal working day, while the DHR has dummies that remove 0.37 GW on a public holiday and 0.56 GW on the Easter days

B4: seasonality in monthly inflation [Proposed]

- ▣ **Question:** is the seasonal pattern of Romanian monthly inflation strong enough to improve one-month forecasts?
- ▣ **Inputs:** HICP, Romania, 2010–2026; $\pi_t = 100 \Delta \ln P_t$ (% per month); model: B3
- ▣ **Tasks**
 1. Regress π_t (2010–2019) on 12 month dummies and test equal month means with an F -test.
 2. Estimate the airline model for $100 \ln P_t$ and report $\hat{\theta}$ and $\hat{\Theta}$.
 3. From monthly origins since January 2016, forecast next month's π with SARIMA, the seasonal naive method and the mean of the last 12 months; compare the MAE.
 4. Test SARIMA against both benchmarks with the DM test (squared errors).
 5. Interpretation: is the seasonal pattern strong enough to improve one-month forecasts?
- ▣ **Report:** an F -test, two estimates, three MAE values, two DM tests and one sentence
- ▣ **Notebook:** section B4

C1: tourism after the pandemic [Proposed]

- **Question:** which training sample gives better forecasts of tourism nights for 2023–2025: data up to 2019, or data up to 2022?
- **Inputs:** Eurostat, nights in tourist accommodation, Romania, monthly, 2005–2025; model: B1
- **Tasks**
 1. Fit the airline model to $\ln y_t$ up to December 2019 and forecast 2023–2025.
 2. Repeat with data up to December 2022.
 3. Compare both with the seasonal naive forecast that repeats 2022, using the MAPE (mean absolute percentage error).
 4. Propose a better treatment of 2020–2021 (intervention dummies, a shorter sample, TBATS or Prophet) and sketch it as a team project.
- **Report:** three MAPE values, the chart and a project plan
- **Notebook:** section C1

C2: audit an AI answer [Proposed]

- A student asked an AI assistant about seasonal models. The answer:
- (a) $\text{SARIMA}(0,1,1)(0,1,1)_4$ has four MA parameters, at lags 1, 4 and 5 plus the noise variance.
- (b) If the ACF of the differenced monthly series has one spike at lag 12 and the PACF decays at 12, 24, 36, use a seasonal $\text{AR}(1)$.
- (c) Unadjusted Romanian GDP fell by about 34% in the first quarter of 2026 compared with the fourth quarter of 2025: a deep recession.
- (d) $\text{DM} = -2.3$ with $d = L(e_1) - L(e_2)$ shows that forecast 2 is significantly more accurate.
- (e) TBATS with periods 7 and 365.25 captures the Easter dip of electricity load.
- (f) The equal-weight average of two unbiased forecasts never has a larger MSE than the average of their two MSEs.
- Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
 - ▶ 2. Report: a list of six verdicts with one line of justification each.

Key takeaways

- ▣ Δ_s removes a stable seasonal pattern; $\Delta\Delta_s$ also removes the trend; the seasonal naive forecast is the benchmark
- ▣ SARIMA multiplies a regular and a seasonal polynomial: the airline model has two parameters and ACF spikes at 1, $s - 1$, s , $s + 1$
- ▣ Calendar effects (working days, Easter, public holidays) are known in advance: put them in as regressors
- ▣ Compare forecasts out of sample, with rolling origins, MASE and the DM test; a visible gain is not always a significant one
- ▣ Combining forecasts is cheap insurance: it is rarely the worst and often near the best

After the seminar

- Lecture 4 derives today's formulas and adds seasonal unit roots, seasonal adjustment (X-13ARIMA-SEATS), multiple seasonality, TBATS and Prophet
- Try the [Proposed] tasks in the notebook; TBATS and Prophet are optional installs there
- C1 can grow into a team project: forecasting tourism or electricity load in Romania with a rolling-origin comparison of several models
- Reading: [Huang and Petukhina \(2022\)](#), Ch. 4–5; [FPP3, Section 9.9](#); [FPP3, Section 5.10](#); [Diebold and Mariano \(1995\)](#)
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

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